

TASK21 - Docker Image Build and Push Report

1. Introduction

Docker is a containerization platform used to package applications along with all required dependencies into a single portable image. This ensures that the application runs consistently across different environments such as development, testing, and production.

In this task, a Docker image of the Driver Drowsiness Detection system was created and pushed to Docker Hub for deployment and sharing purposes.

2. Docker Image Creation

The Docker image was created using the following command:

```
docker build -t driver-drowsiness .
```

This command reads the Dockerfile, installs required dependencies, and packages the application into a Docker image.

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

D:\P\mlops\TASKS\DAY - 10\driver_drowsiness> docker build -t driver-drowsiness .
[+] Building 3.2s (12/12) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 373B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 2.4s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 268B 0.0s
=> CACHED [2/6] WORKDIR /app 0.0s
=> CACHED [3/6] RUN apt-get update && apt-get install -y libgl1 0.0s
=> CACHED [4/6] COPY requirements.txt . 0.0s
=> CACHED [5/6] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [6/6] COPY . . 0.0s
=> exporting image 0.6s
=> => exporting layers 0.3s
=> => exporting manifest sha256:eeb290d40ecc5599fcc31a1f0f6422ed420 0.0s
=> => exporting config sha256:198c4a9ec8191a99d090fb68aff201bea72480 0.0s
=> => exporting attestation manifest sha256:bee8bf9ad779a61b604774 0.0s
=> => exporting manifest list sha256:c0da6968f96cc7b0d5d1356862dd9b6 0.0s
=> => naming to docker.io/library/driver-drowsiness:latest 0.0s
=> => unpacking to docker.io/library/driver-drowsiness:latest 0.1s

D:\P\mlops\TASKS\DAY - 10\driver_drowsiness> docker run -p 5000:5000 driver-drowsiness
WARNING: user config directory '/root/.config/Ultralytics' is not writable, using '/tmp/Ultralytics'. Set YOLO_CONFIG_DIR to override.
Creating new Ultralytics Settings v0.0.6 file
View Ultralytics Settings with 'yolo settings' or at '/tmp/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/quickstart/#ultralytics-settings.
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
```

3. Docker Image Tagging

After building the image, it was tagged to match the Docker Hub repository:

```
docker tag driver-drowsiness gopikaas/driver-drowsiness:latest
```

This step links the local image with the remote Docker Hub repository.

4. Docker Login

To push the image to Docker Hub, authentication was performed using:

```
D:\P\mlops\TASKS\DAY - 10\driver_drowsiness> docker login
Authenticating with existing credentials... [Username: gopikaas]

Info → To login with a different account, run 'docker logout' followed by
'docker login'

Login Succeeded
```

The system successfully authenticated the user.

5. Pushing Image to Docker Hub

The image was pushed using the following command:

```
docker push gopikaas/driver-drowsiness:latest
```

```
D:\P\mlops\TASKS\DAY - 10\driver_drowsiness>docker push gopikaas/driver-drowsiness:latest
The push refers to repository [docker.io/gopikaas/driver-drowsiness]
a9cf18ddfb28: Already exists
a66698a8a5b9: Layer already exists
3581f9a385a6: Layer already exists
8649771fee17: Layer already exists
0b2f22071300: Layer already exists
37522a1c6a83: Pushed
45006ceeeea9: Layer already exists
acf65e32ab2e: Layer already exists
5b4d6ff92fc4: Layer already exists
797d495f2c68: Layer already exists
latest: digest: sha256:4d6f1e598ec8326d682ee301fa51df716cf8e98b0313bbbf62f053a679039ea5 size: 856
```

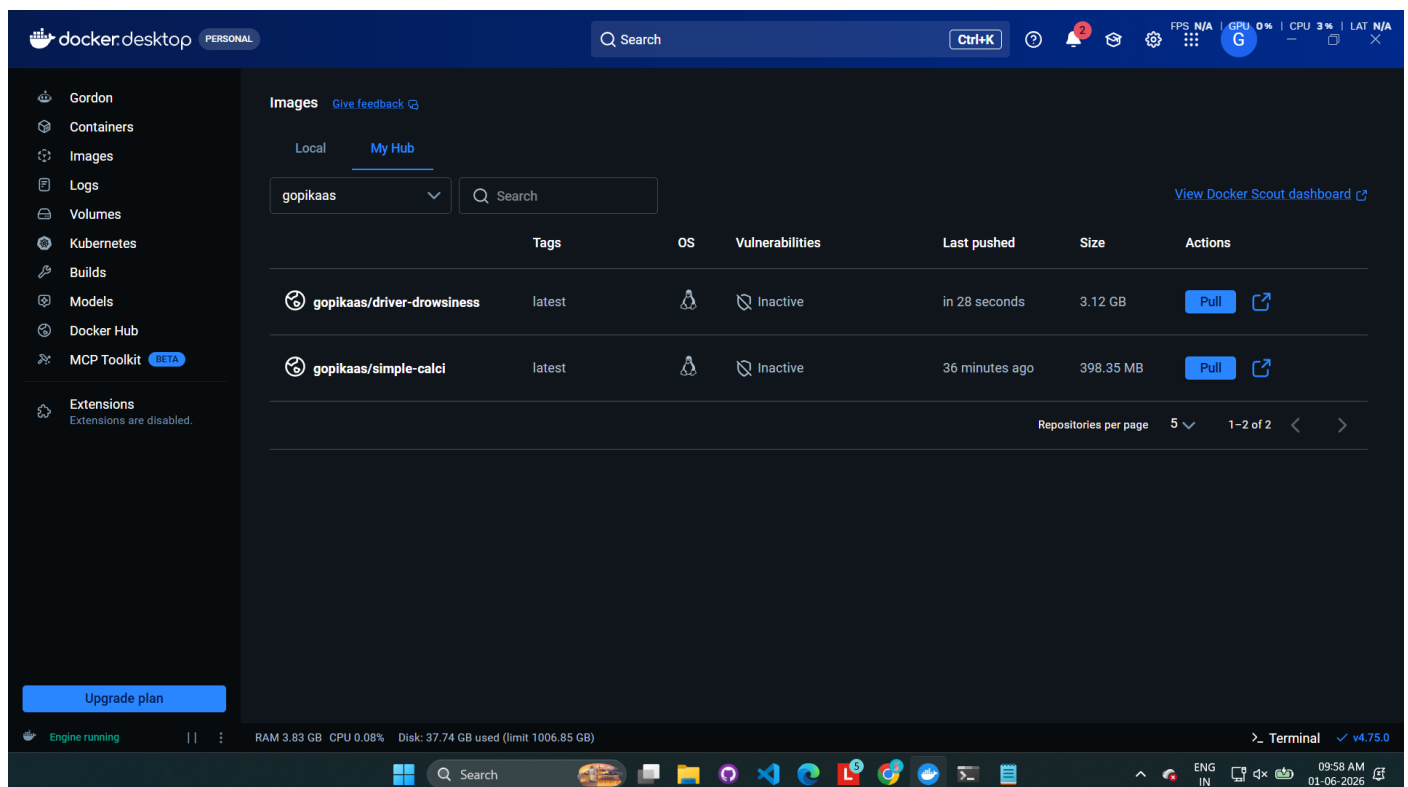
During the push process, Docker uploaded image layers to the repository. Some layers were already present and reused, while others were uploaded successfully.

6. Docker Hub Repository

The image is now available in the Docker Hub repository:

Repository Name:

gopikaas/driver-drowsiness



8. Result

The Docker image was successfully built, tagged, authenticated, and pushed to Docker Hub. The container is now ready for deployment on any system that supports Docker.